A Probabilistic Construction for the Transcendental Form of Erdős Problem 906

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Abstract

Erdős Problem 906 asks whether there is a non-zero entire function f such that, for every infinite increasing sequence of positive integers $n_1 < n_2 < \cdots$, the union of the zero sets of the derivatives $f^{(n_k)}$ is dense in $\mathbb{C}$. The literal statement is trivial if polynomials are allowed. This note addresses the non-trivial interpretation in which f is required to be transcendental entire. We give a probabilistic construction using the planar Gaussian entire function

$$F(z) = \sum_{k=0}^{\infty} \xi_k \frac{z^k}{\sqrt{k!}},$$

where the ξ_k are independent standard complex Gaussian random variables. For the n-th derivative $F^{(n)}$, the Edelman–Kostlan formula and the Plancherel–Rotach asymptotic for Laguerre polynomials imply that, on every compact set avoiding the origin, the expected zero measure grows at least on the order of $\sqrt{n}$. An Offord-type deviation estimate for linear statistics of zeros of Gaussian analytic functions then gives a summable bound for the probability that $F^{(n)}$ has no zero in a fixed disk. Borel–Cantelli, applied to a countable base of rational disks whose closures avoid the origin, yields a realization of F for which every non-empty open set contains zeros of all sufficiently high derivatives. Since F is transcendental entire almost surely, this proves existence for the transcendental version of the problem, subject to the standard estimates cited below.

1 The problem and a cofinite reformulation

For an entire function f, write

$$Z_n(f) = \{z \in \mathbb{C} : f^{(n)}(z) = 0\}.$$

The literal version of Erdős Problem 906 is immediate if polynomials are admitted: if f is a non-zero polynomial, then $f^{(n)} \equiv 0$ for all sufficiently large n, so every sufficiently high derivative has zero set equal to all of $\mathbb{C}$. Thus the interesting form of the question is the transcendental one.

Lemma 1 (Cofinite hitting property)**.** *For an entire function f, the following are equivalent.*

(i) *For every infinite increasing sequence $n_1 < n_2 < \cdots$, the set $\bigcup_{k\geq 1} Z_{n_k}(f)$ is dense in $\mathbb{C}$.*

(ii) *For every non-empty open set $U \subset \mathbb{C}$, there is an integer $N(U)$ such that $Z_n(f) \cap U \neq \varnothing$ for every $n \geq N(U)$.*

Proof. Assume (ii). Let $n_1 < n_2 < \cdots$ be any infinite increasing sequence and let U be a non-empty open set. Since $n_k \to \infty$, some n_k is at least $N(U)$; hence $Z_{n_k}(f) \cap U \neq \varnothing$. Therefore the union in (i) meets every non-empty open set and is dense.

Conversely, if (ii) fails, then some non-empty open set U is missed by $Z_n(f)$ for infinitely many n. Listing those integers increasingly as $n_1 < n_2 < \cdots$, the union $\bigcup_k Z_{n_k}(f)$ is disjoint from U, and hence is not dense. This proves the equivalence. □

It remains to construct a transcendental entire function satisfying the cofinite hitting property.

2 The planar Gaussian entire function

Let

$$F(z) = \sum_{k=0}^{\infty} \xi_k \frac{z^k}{\sqrt{k!}},$$

where $\xi_0, \xi_1, \ldots$ are independent standard complex Gaussian random variables. This is the planar Gaussian entire function. It is entire almost surely, and it is not a polynomial almost surely, since the probability that all but finitely many of the independent Gaussian coefficients vanish is zero.

For each $n \geq 0$, $F^{(n)}$ is a Gaussian analytic function with covariance kernel

$$K_n(z,w) = \mathbb{E}\{F^{(n)}(z)\overline{F^{(n)}(w)}\} = \partial_z^n \partial_{\overline{w}}^n e^{z\overline{w}}.$$

On the diagonal this kernel has the explicit form

$$K_n(z,z) = n! e^{|z|^2} L_n(-|z|^2), \tag{1}$$

where $L_n = L_n^{(0)}$ denotes the Laguerre polynomial. Indeed, expanding $e^{z\overline{w}}$ and differentiating gives

$$K_n(z,z) = \sum_{j=0}^{\infty} \frac{(n+j)!}{(j!)^2} |z|^{2j} = n! \sum_{j=0}^{\infty} \binom{n+j}{n} \frac{|z|^{2j}}{j!},$$

which is equivalent to (1) by the standard generating identity

$$e^x L_n(-x) = \sum_{j=0}^{\infty} \binom{n+j}{n} \frac{x^j}{j!}.$$

By the Edelman–Kostlan formula, the expected zero counting measure $\mu_n = \mathbb{E}[n_{F^{(n)}}]$ is

$$d\mu_n(z) = \frac{1}{4\pi} \Delta \log K_n(z,z)\, dA(z), \tag{2}$$

where Δ is the usual Euclidean Laplacian and dA is planar Lebesgue measure.

3 Growth of the expected zero measure

Lemma 2 (Expected mass in disks away from the origin)**.** *Let $D \subset \mathbb{C}$ be a disk whose closure does not contain 0. Let $\varphi \in C_c^2(D)$ be non-negative and not identically zero. Then there are constants $c_\varphi > 0$ and n_0 such that, for all $n \geq n_0$,*

$$\mathbb{E}\left[\int \varphi\, dn_{F^{(n)}}\right] = \int \varphi\, d\mu_n \geq c_\varphi \sqrt{n}.$$

Proof. Since $\operatorname{supp} \varphi$ is compact and avoids the origin, there are numbers $0 < r_0 < r_1 < \infty$ such that

$$r_0 \leq |z| \leq r_1 \qquad (z \in \operatorname{supp} \varphi).$$

For x in compact subintervals of $(0, \infty)$, the Plancherel–Rotach asymptotic for Laguerre polynomials gives, uniformly in x,

$$L_n(-x) = \frac{1 + o(1)}{2\sqrt{\pi}}\, e^{-x/2} x^{-1/4} n^{-1/4} e^{2\sqrt{nx}}. \tag{3}$$

Thus, with $x = |z|^2$ and uniformly on $\operatorname{supp}\varphi$,

$$\log L_n(-|z|^2) = 2\sqrt{n}\,|z| - \frac{|z|^2}{2} + O(\log n) + O(1),$$

where the $O(\log n)$ term is independent of z up to harmless constants on the fixed annulus. Combining this with (1), we obtain

$$\log K_n(z,z) = C_n + 2\sqrt{n}\,|z| + \frac{|z|^2}{2} + O(1), \tag{4}$$

uniformly on $\operatorname{supp}\varphi$, where C_n is independent of z.

Using (2) in distribution form,

$$\int \varphi\, d\mu_n = \frac{1}{4\pi}\int_{\mathbb{C}} \log K_n(z,z)\Delta\varphi(z)\, dA(z). \tag{5}$$

The term C_n contributes nothing because $\int \Delta\varphi\, dA = 0$. The bounded terms contribute $O(1)$. The leading term gives

$$\frac{\sqrt{n}}{2\pi}\int_{\mathbb{C}} |z|\Delta\varphi(z)\, dA(z).$$

By integration by parts, since $\operatorname{supp}\varphi$ avoids 0,

$$\int_{\mathbb{C}} |z|\Delta\varphi(z)\, dA(z) = \int_{\mathbb{C}} \varphi(z)\Delta|z|\, dA(z) = \int_{\mathbb{C}} \frac{\varphi(z)}{|z|}\, dA(z) > 0.$$

Therefore (5) is at least $c_\varphi\sqrt{n}$ for all sufficiently large n. □

4 Offord estimate and summability of hole probabilities

We use the following standard Offord-type estimate for zero statistics of Gaussian analytic functions. If g is a Gaussian analytic function, $\mu_g = \mathbb{E}[n_g]$, and $\varphi \in C_c^2(\mathbb{C})$, then for every $\lambda > 0$,

$$\mathbb{P}\left\{\left|\int \varphi\,(dn_g - d\mu_g)\right| \geq \lambda\right\} \leq 3\exp\left(-c\frac{\lambda}{\|\Delta\varphi\|_{L^1}}\right), \tag{6}$$

with an absolute positive constant c. This is the usual Offord–Sodin estimate for linear statistics of zeros of Gaussian analytic functions.

Lemma 3 (Summable hole probabilities)**.** *Let $D \subset \mathbb{C}$ be a disk whose closure does not contain 0. Then there are constants $C_D, c_D > 0$ and n_0 such that, for all $n \geq n_0$,*

$$\mathbb{P}\{F^{(n)} \textit{ has no zero in } D\} \leq C_D e^{-c_D\sqrt{n}}.$$

Consequently,

$$\sum_{n=1}^{\infty} \mathbb{P}\{F^{(n)} \textit{ has no zero in } D\} < \infty.$$

Proof. Choose $\varphi \in C_c^2(D)$ with $0 \leq \varphi \leq 1$ and φ not identically zero. By Lemma 2,

$$\int \varphi\, d\mu_n \geq c_\varphi\sqrt{n}$$

for all large n.

If $F^{(n)}$ has no zero in D, then $\int \varphi\, dn_{F^{(n)}} = 0$. Hence

$$\left|\int \varphi\,(dn_{F^{(n)}} - d\mu_n)\right| \geq c_\varphi\sqrt{n}.$$

Applying (6) to $g = F^{(n)}$ gives

$$\mathbb{P}\{F^{(n)} \text{ has no zero in } D\} \leq 3\exp\left(-c\frac{c_\varphi\sqrt{n}}{\|\Delta\varphi\|_{L^1}}\right),$$

which is of the required form. Since $\sum_n e^{-c\sqrt{n}} < \infty$, the summability follows. □

5 Conclusion

Theorem 1. *There exists a transcendental entire function f such that, for every infinite increasing sequence of positive integers $n_1 < n_2 < \cdots$, the set*

$$\bigcup_{k\geq 1}\{z \in \mathbb{C} : f^{(n_k)}(z) = 0\}$$

is dense in $\mathbb{C}$.

Proof. Let $\mathcal{B}$ be a countable base of disks with rational centers and rational radii whose closures avoid 0. This is still a base for the density problem: every non-empty open set in $\mathbb{C}$ contains at least one disk from $\mathcal{B}$. Indeed, if the open set contains a non-zero point, choose a sufficiently small rational disk around it; if it contains the origin, openness supplies nearby non-zero points and the same argument applies.

Fix $D \in \mathcal{B}$. By Lemma 3 and the Borel–Cantelli lemma, with probability one only finitely many events

$$\{F^{(n)} \text{ has no zero in } D\}$$

occur. Hence, with probability one, there is $N(D)$ such that $F^{(n)}$ has a zero in D for every $n \geq N(D)$.

Since $\mathcal{B}$ is countable, the intersection of these probability-one events over all $D \in \mathcal{B}$ still has probability one. Therefore there exists a realization f of the planar Gaussian entire function such that, for every $D \in \mathcal{B}$, all sufficiently high derivatives $f^{(n)}$ have a zero in D.

Let $U \subset \mathbb{C}$ be any non-empty open set. Choose $D \in \mathcal{B}$ with $D \subset U$. Then, for all sufficiently large n, $f^{(n)}$ has a zero in D, hence in U. By Lemma 1, the union of the zero sets of the derivatives along any infinite subsequence is dense in $\mathbb{C}$.

Finally, f is transcendental entire almost surely, as noted above. Thus at least one realization has all the required properties. □

Remark 1. *The proof is existential and probabilistic. It does not give an explicit deterministic formula for a particular function f, but almost every realization of the planar Gaussian entire function has the required property.*

Remark 2. *The only place where disks avoiding 0 are used is the Laguerre asymptotic and the identity $\Delta|z| = 1/|z|$ away from the origin. This causes no loss for the original density problem, because every non-empty open subset of $\mathbb{C}$ contains a smaller disk whose closure avoids 0.*

References

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